



Attack Surface Management – Team 1

Team 1 Objective :

Continuously discover and inventory internet-facing assets.

Discovery Pipeline :

Subdomain Discovery



DNS Resolution



Port Scanning



Technology Detection



Threat Intel Enrichment

Technologies Used :

- Python
- FastAPI
- PostgreSQL
- Shodan
- Censys
- Httpx



- Naabu
- Nmap

High-Level Architecture :

Target Domain Input



Subdomain Enumeration



DNS Resolution



Port Discovery



HTTP Tech Detection



Threat Intel Enrichment

(Shodan/Censys/VT)



Data Normalization



PostgreSQL DB



FastAPI APIs



Step 1 — Setup Python Environment

Step 2 — Install Recon Tools (Subfinder, Httpx, Naabu, Nmap)

PHASE 1 — Subdomain Discovery

First we identify all externally exposed domains.

Step 3 — Build Subdomain Scanner

PHASE 2 — DNS & IP Resolution

Once domains are discovered, we resolve them into infrastructure IPs.

Step 4 — Create DNS Resolver

Step 5 — Combine Discovery + Resolution

PHASE 3 — Port Scanning

identify exposed services and attack vectors.

Step 6 — Integrate Naabu

PHASE 4 — HTTP Technology Detection

Step 7 — Integrate Httpx

PHASE 5 — Threat Intelligence Enrichment

Step 8 — Integrate Shodan

Step 9 — Integrate Censys



PHASE 6 — Data Normalization

All collected information is standardized into a unified schema.

Step 10 — Create Unified Asset Schema

PHASE 7 — Database Storage

Step 11 — PostgreSQL Tables

PHASE 8 — API Layer

Step 12 — FastAPI Integration